Simple Risk Engine v1.1

1. INTRODUCTION

1.1. Objective

To upgrade from a minimum viable product (MVP) of Simple Risk Engine v1.1.4 by implementing faster, safer, and more efficient system using HTMX + FastAPI combination. Optimization also implemented in some areas.

1.2. Improvements

- 1.1.5. Added volatility logic using atr (atr_c) and slippage logic (est_slippage)
- 1.1.5. Change streamlit frontend to HTMX

1.3. Libraries

1.3.1. Data and Exchange

- ccxtpro or ccxt
- requests
- websocket
- python-dotenv: to secure api key and api secret

1.3.2. Logic and Data Logging

- pandas
- numpy
- asyncio
- openpyxl

1.3.3. Time

- time
- schedule

1.3.4. Local Webhost Dashboard

- Frontend: HTMX
- Backend: FastAPI + WebSocket Connection or exchange API

1.4. Configs

- Exchange: Binance
- Data Pooling: Binance
- Market: USD-M Futures (Crypto)
- Initial Equity: existing real-time equity on binance
- Timezone: UTC +7

1.5. Approach

1.5.1. Risk Management

Not using Kelly Criterion, only:

- Individual position based on risk per trade and stop-loss percentage
- Added volatility logic using Average True Range (real-time)
- Added slippage logic using VWAP (real-time)
- Warning and hard stops of maximum loss of a day, week and month.
- Correlation risks to determine combined entire portfolio exposure.

1.5.2. Trade and Risk History

- Real-time account data
- Daily portfolio dashboard snapshots converted into .csv file at beginning of the day (_bod).
- Monthly portfolio dashboard snapshots converted into .csv file, reset at first day on the month at _bod.

1.5.3. Local Web Host Output

Components are:

- Portfolio Dashboard (real-time)
- Entry Risk Calculator and Estimator (real-time data fetching)
- Trade and Risk History (real-time from last 30 days)
- Editable Parameters

2. FRAMEWORK

2.1. Data Pooling & Processing

- Exchange's API (view-only) for real-time portfolio updates, snapshot stored in risk journal periodically e.g., every 00.00 utc+7

- All real-time data are using WebSocket or binance fapi (futures API) connection to prevent constant API calls which can messes up with exchange API rate limit

2.2. Features

2.2.1. Dashboard of Current Portfolio Exposure and Risk

Output = real-time portfolio_dashboard on local webhost

- exchange_info = *real-time* information related to exchange used such as exchange name, account ID, server, latency, fees, etc.
- account_state = *real-time* current information such as balances, current pnl, equity, available margin, beginning of the day (_bod), start of week (_sow), cashflows, total_position_value_usdt, total_margin_ratio , total_margin_used, total_unrealized, total_realized, total_tp_usdt, total_sl_usdt, daily_unrealized, daily_realized, daily_pnl, daily_pnl_percent, total_equity, max_total_equity (snapshot and reset every _bod), min_total_equity, etc
- position_info = *real-time* data related to certain open positions, such as ticker, order_timestamp, entry_timestamp, contract_amount, contract_size, direction, position_value_usdt, position_value_asset, average, fair_price , liquidation_price, individual_margin_ratio, individual_margin_used, individual_unrealized, individual_funding_fees, individual_realized, individual_tpsl, individual_tp_price, individual_sl_price, individual_tp_amount, individual_sl_amount, individual_tp_usdt, individual_sl_usdt, and model_name = model name used from risk calculator, add suffix to every position info from earliest position e.g., position_info1, position_info2 and so on.
- calc_portfolio_stats = *real-time* overall portfolio calculations such as total_exposure, total_weekly_pnl, total_weekly_pnl_percent, total_correlated_exposure, and _dd = *real-time* drawdown calculated by (max_equity - current_equity) / max_equity
- _params = all parameter used in risk engine which are individual_risk_per_trade, max_w_loss_percent = editable maximum accumulated weekly loss percent, reset every sow bod, dd_baseline_equity (a baseline that reset every _bod), max_dd_percent, max_exposure = editable maximum calc_exposure + potential_exposure to enter a new position, and max_ position_count, all aspect on dashboard are view only

note: all data to calculate using exchange's API via websocket to display real-time portfolio data

2.2.2. Entry Risk Calculator

Output = real-time risk_calculator on local webhost

- ticker = input
- average = input
- one_percent_depth = *real-time* liquidity around 1% of entry price
- current_orderbook = *real-time* current orderbook
- individual_risk = from parameter, not editable in risk calculator
- individual_tpsl
  - individual_tp_price = input
  - individual_tp_amount = 1 to 100% position
  - individual_sl_price = input
  - individual_sl_amount = 1 to 100% position
- _side = calculated automatically using SL, if SL are above average inputted, then its short, vice versa.
- risk_usdt = individual_risk * total_equity (*real-time*)
- atr_c = (*real-time*) ATR (100, 4h) / ATR (14, 4h), capped at 1
  - if 0.2 => atr_c > 0.6 then categorize as volatile (yellow colored)
  - if 0.6 => atr_c > 1 then categorize as normal (green colored)
  - if atr_c => 1 then categorize as not volatile (green colored)
  - if atr_c < 0.2 then categorize as too volatile (red colored) and display volatility exceeds maximum threshold, so particular trade is ineligible.
- effective_entry = (*real-time*) 1 – est_slippage
- _size = (*real-time*) effective_entry * atr_c * risk_usdt / (| individual_sl_price – average | / average)
- correlated_sectors = group the market correlations between assets
  - big_two_crypto = bitcoin and Ethereum, BTCUSDT and ETHUSDT
  - top_twenty_alts = higher beta than big_two_crypto, top 20 coins by market cap excluding bitcoin, Ethereum, and stablecoins
  - other_alts = other altcoins
  - commodities = gold and silver, XAUUSDT and XAGUSDT
- correlated_exposure = (*real-time*) sum of long and shorts notional size for each correlated_sectors, long is positive size, short is negative size. Each

sector cannot exceeds max_correlated_exposure, else then display a limit where new position cannot be added instead of size and estimations.

- estimations = group of estimations
  - est_slippage = *real-time* |VWAP_fill - average| / average. Also add a disclaimer label in the UI that slippage estimates are indicative only.
  - est_slippage_usdt = est_slippage * position_value_usdt
  - est_profit = individual_tp_usdt – 2 * maker_fee – 2 *est_slippage_usdt
  - est_loss = individual_sl_usdt – 2 * maker_fee – 2 * est_slippage_usdt
  - est_r = estimation of risk to reward ratio calculated by est_profit / est_loss
  - est_exposure = estimated potential exposure (total_notional_size + size (from calculated potential position)) / total_equity
  - est_correlated_exposure = display total_correlated_exposure and correlated_sector addition
- max_position_count = if true then don't display size and estimations, display a trading limit instead, else ignore
- weekly_loss_warning and max_dd_warning = if true then display warning message, else ignore
- weekly_loss_limit and max_dd_limit = if true then don't display size and estimations, display a trading limit instead, else ignore

2.2.3. Trade and Risk History

Output = real-time data viewing on local webhost and exported .csv periodically

- portfolio_dashboard_snapshots = periodic snapshots of dashboard data, daily and monthly including exchange_info, account_state, position_info, calc_portfolio_stats, weekly_pnl_state, dd_state and _params

- pre_trade_log = *real-time update* showing data including cancelled potential trades
  - ticker
  - average
  - one_percent_depth
  - current_orderbook
  - individual_risk
  - individual_tpsl
  - individual_tp_price
  - individual_tp_amount

    - individual_tp_usdt
    - individual_sl_price
    - individual_sl_amount
    - individual_sl_usdt
    - model_name
    - model_desc
    - risk_usdt
    - size
- execution_log = *real-time* showing execution data, monthly
    - entry_timestamp = timestamp when entry trade executed
    - entry_price_actual = actual entry price compared to plan at risk calculator
    - size_filled = partial or full based on plan
    - slippage
    - order_type = limit or market order
    - maker_fee
    - taker_fee
    - latency_snapshot = latency when executing
    - orderbook_depth_snapshot = depth when executing
- live_trades_log = *real-time* live trades data, monthly
    - max_profit = maximum unrealized profit before closing
    - max_loss = maximum unrealized loss before closing
    - hold_time = time from entry, format [ddd ,hh ,mm , ss]
    - stop_adjustments = true or false, if true then display changes and timestamp
- trade_history = *real-time* closed trades, monthly
    - exit_timestamp = timestamp when exit trade executed
    - exit_price = actual exit price compared to plan at risk calculator
    - notes = input if any condition occured
    - individual_realized
    - individual_realized_r
    - total_funding_fees = total funding paid and earned
    - total_fees = total exchange fee paid
    - slippage_exit = slippage at exit compared to planned
    - holding_time = time from entry, format [ddd ,hh ,mm , ss]
- exported_algorithm = export data to excel in .csv format periodically

- manual_export = manual export button

  note: always load the last 30 days trade and risk history data both from exchange and risk calculator

2.2.4. Params

Output = editable params on localwebhost

- _params = all editable parameter used in risk engine which are individual_risk_per_trade, max_w_loss_percent = editable maximum accumulated weekly loss percent, reset every Monday _bod, max_dd_percent, max_exposure = editable maximum calc_exposure + potential_exposure to enter a new position, and max_ position_count
- soft_hard_stops = all editable warning and hardstops for weekly limit and drawdown limits such as total_equity, total_weekly_pnl, max_w_loss, weekly_loss_warning = show a warning in dashboard and risk calculator if total_weekly_pnl / total_equity at sow equals to 80%, weekly_loss_limit = show a hard warning in dashboard and risk calculator if total_weekly_pnl / total_equity at sow equals to 95%, _dd, max_dd_percent, max_dd_warning = show a warning in dashboard and risk calculator if dd / max_dd_percent equals to 80%, max_dd_limit = show a hard warning in dashboard and risk calculator if dd / max_dd_percent equals to 95%. Outputs are weekly_pnl_state and dd_state.
- auto_export = editable automatic export period
- max_correlated_exposure = editable max sum of correlated exposure, default total correlated exposure is 0.5 * total_equity
- _api = api key and secret from .env, api key only shows the first 5 letter and the rest is on asterix.

2.2.5. Websocket Related

Output = display system latency, error, packet losses, retries and solving log on top right corner part of the web

- _latency = Display a heartbeat/ping interval,
- _reconnections = reconnections retry policy with backoff and retries attemps
- packet_loss = a "data staleness" indicator in the dashboard that shows time since last WebSocket update, and (d) fallback to REST polling if WebSocket has been disconnected for more than 30 seconds.

- ws_log = all connection related log, only display last 10 logs

  Note: All real-time data only refreshes the data not overall webpage

2.2.6. Local Web Host Shortcut or Launcher

- Added a shortcut to automatically deploy local webhost

3. DEBUGGING & RECALIBRATIONS HISTORY

3.1. Version 1.1

3.1.1. [Major] Real-time data using WebSocket system to prevent exchange rate limit

3.1.2. [Major] Instead of python_binance, exchange library now using CCXT (Cryptocurrency Exchange Trading Library) for flexible exchange options

3.1.3. [Major] Using Asyncio for faster data streams

3.1.4. [Major] Main risk engine switches to 3 different sections, Portfolio Dashboard, Risk Calculator, and Editable Parameters. Warnings and hard stops moved to dashboard section.

3.1.5. [Major] architectural changes in framework

3.1.6. Changed variables into definitions in the framework

3.1.7. Fixing a bug where TPSL doesn’t update regularly

3.1.8. Shortcut .bat app to instantly open localhost website

3.1.9. Fixing a bug where data is not showed properly e.g., balance showing 0